

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

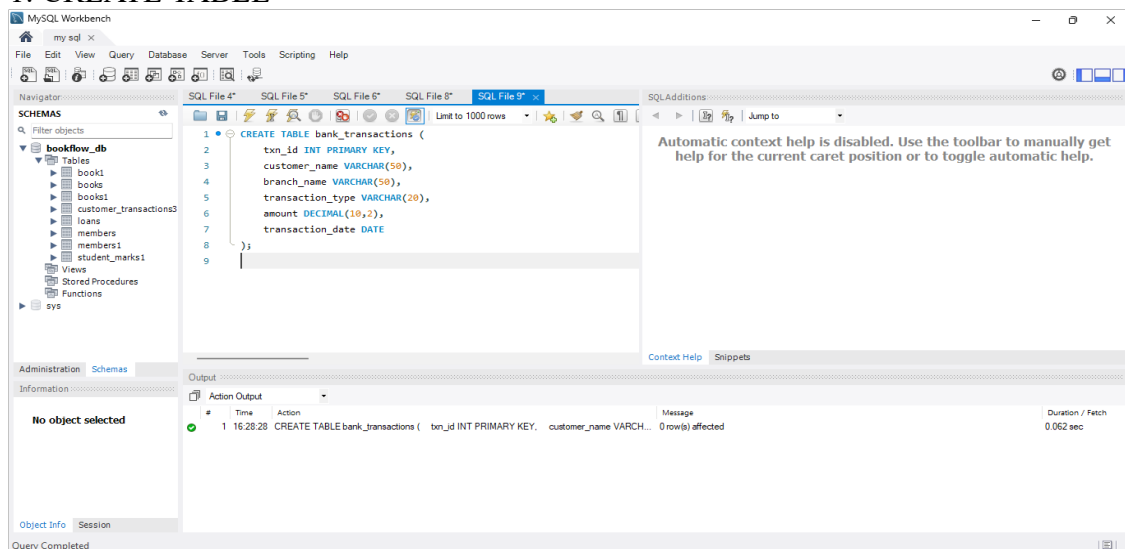
Sec:- 3

Name:- Srithika Kambhammettu

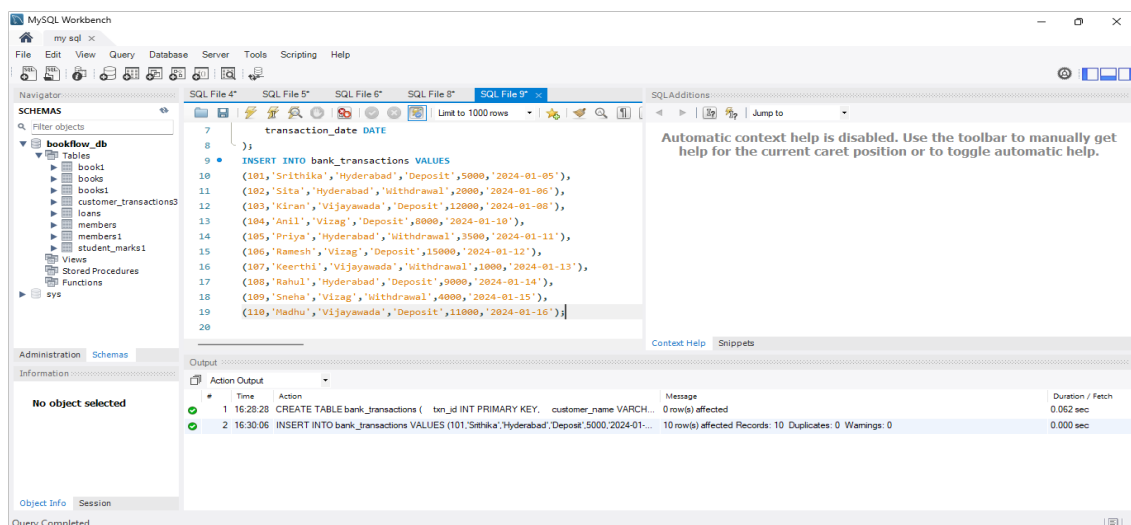
Roll no:- 2520030361

3- Aggregate Functions & SQL Clauses

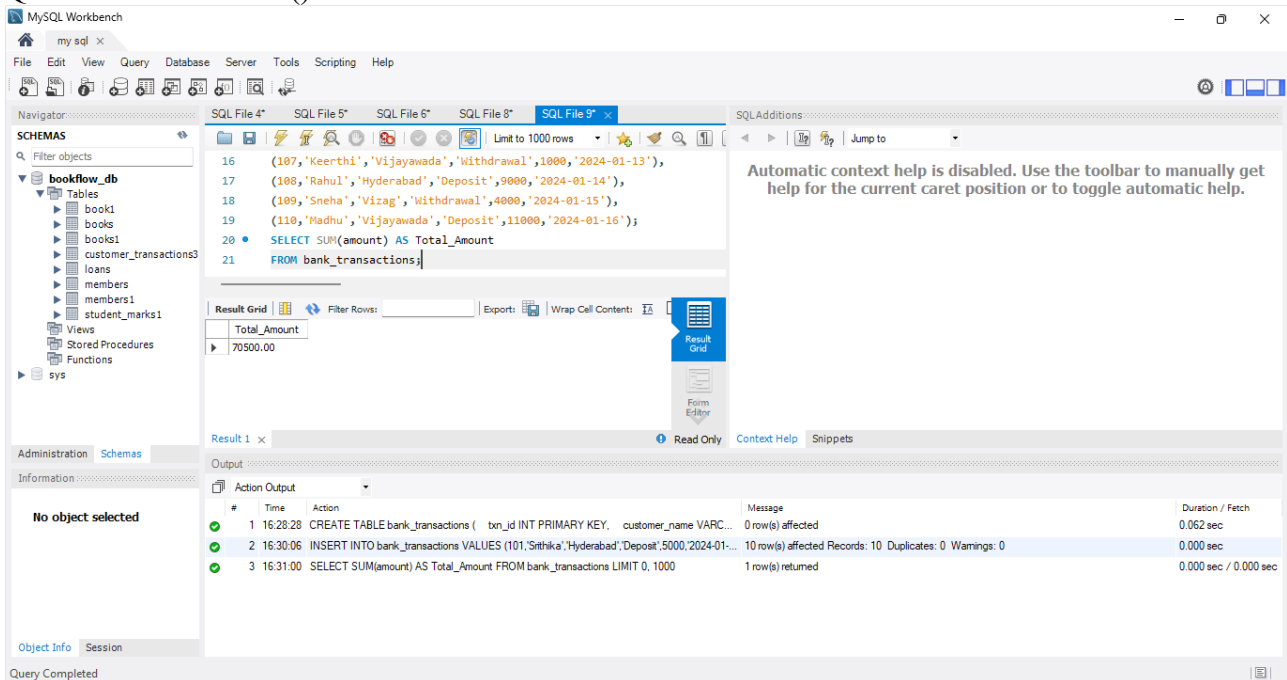
STEP 1: CREATE TABLE



STEP 2: INSERT DATA



QUESTION 1 – SUM()



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' panel with a tree view of the 'bookflow_db' database, including tables like 'book1', 'books', 'books1', 'customer_transactions3', 'loans', 'members', 'members1', 'student_marks1', 'Views', 'Stored Procedures', and 'Functions'. The main editor window shows a SQL script with the following queries:

```
16 (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
17 (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
18 (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
19 (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
20 • SELECT SUM(amount) AS Total_Amount
21 FROM bank_transactions;
```

The 'Result Grid' shows the output of the SUM query:

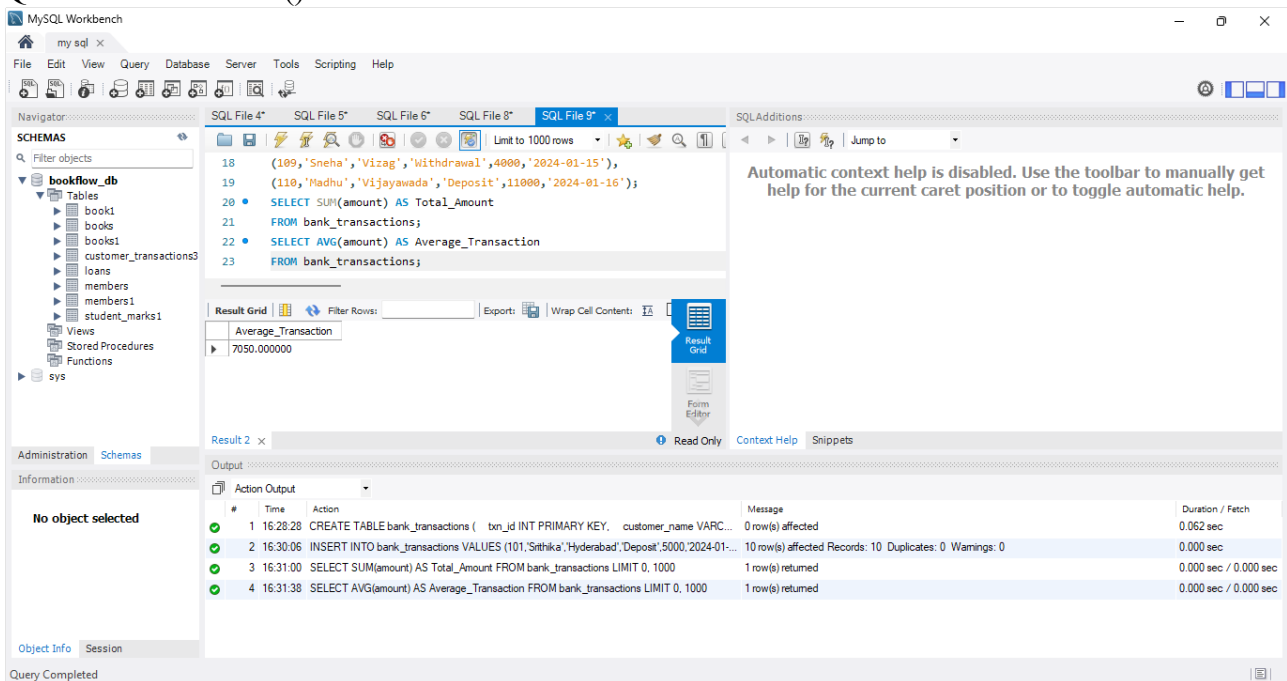
Total_Amount
70500.00

The 'Output' panel shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	16:28:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(100), amount INT, txn_date DATE);	0 row(s) affected	0.062 sec
2	16:30:06	INSERT INTO bank_transactions VALUES (101,'Sithika','Hyderabad','Deposit',5000,'2024-01-13');	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	16:31:00	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The status bar at the bottom indicates 'Query Completed'.

QUESTION 2 – AVG()



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'Schemas' panel with a tree view of the 'bookflow_db' database. The main editor window shows a SQL script with the following queries:

```
18 (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
19 (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
20 • SELECT SUM(amount) AS Total_Amount
21 FROM bank_transactions;
22 • SELECT AVG(amount) AS Average_Transaction
23 FROM bank_transactions;
```

The 'Result Grid' shows the output of the AVG query:

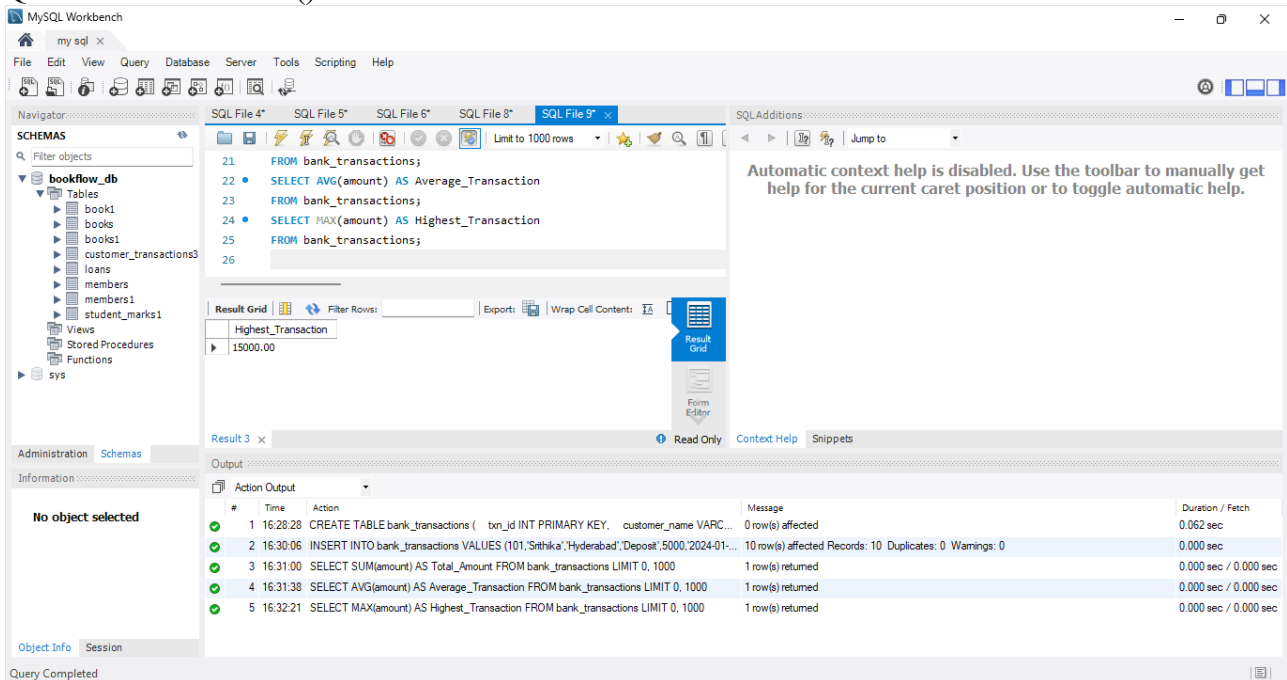
Average_Transaction
7050.000000

The 'Output' panel shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	16:28:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(100), amount INT, txn_date DATE);	0 row(s) affected	0.062 sec
2	16:30:06	INSERT INTO bank_transactions VALUES (101,'Sithika','Hyderabad','Deposit',5000,'2024-01-13');	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	16:31:00	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
4	16:31:38	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The status bar at the bottom indicates 'Query Completed'.

QUESTION 3 – MAX()



The screenshot shows MySQL Workbench with a query window containing SQL code for Question 3. The query is as follows:

```
21 FROM bank_transactions;
22 • SELECT AVG(amount) AS Average_Transaction
23 FROM bank_transactions;
24 • SELECT MAX(amount) AS Highest_Transaction
25 FROM bank_transactions;
26
```

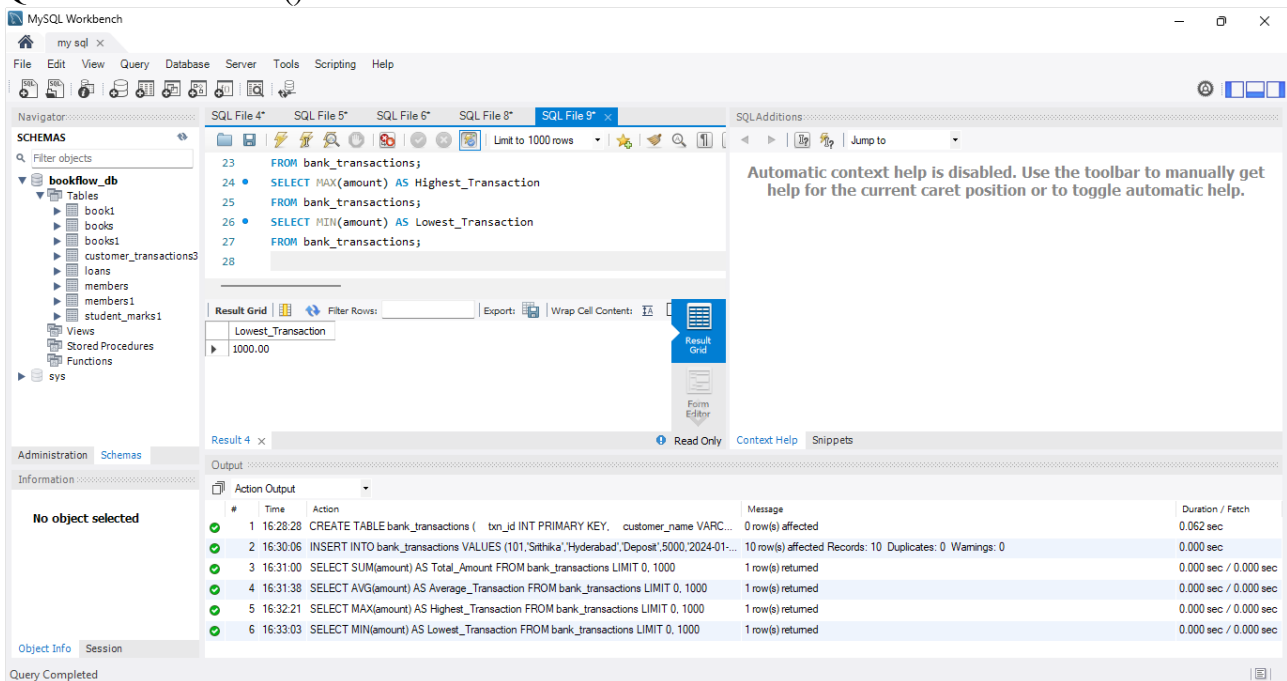
The result grid shows the output for the query:

	Highest_Transaction
1	15000.00

The output pane shows the following messages:

#	Time	Action	Message	Duration / Fetch
1	16:28:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(100), amount DECIMAL(10,2), txn_date DATE)	0 row(s) affected	0.062 sec
2	16:30:06	INSERT INTO bank_transactions VALUES (101,'Sithika','Hyderabad','Deposit',5000,'2024-01-01')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	16:31:00	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
4	16:31:38	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	16:32:21	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

QUESTION 4 – MIN()



The screenshot shows MySQL Workbench with a query window containing SQL code for Question 4. The query is as follows:

```
23 FROM bank_transactions;
24 • SELECT MAX(amount) AS Highest_Transaction
25 FROM bank_transactions;
26 • SELECT MIN(amount) AS Lowest_Transaction
27 FROM bank_transactions;
28
```

The result grid shows the output for the query:

	Lowest_Transaction
1	1000.00

The output pane shows the following messages:

#	Time	Action	Message	Duration / Fetch
1	16:28:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(100), amount DECIMAL(10,2), txn_date DATE)	0 row(s) affected	0.062 sec
2	16:30:06	INSERT INTO bank_transactions VALUES (101,'Sithika','Hyderabad','Deposit',5000,'2024-01-01')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	16:31:00	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
4	16:31:38	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	16:32:21	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:33:03	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

QUESTION 5 – COUNT()

The screenshot shows MySQL Workbench with SQL File 9 open. The query is as follows:

```
24 • SELECT MAX(amount) AS Highest_Transaction
25 • FROM bank_transactions;
26 • SELECT MIN(amount) AS Lowest_Transaction
27 • FROM bank_transactions;
28 • SELECT COUNT(*) AS Total_Transactions
29 • FROM bank_transactions;
```

The Result Grid shows the output of the query:

	Total_Transactions
10	

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
2	16:30:06	INSERT INTO bank_transactions VALUES (101,'Sithika','Hyderabad','Deposit',5000,2024-0...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	16:31:00	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
4	16:31:38	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	16:32:21	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:33:03	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	16:33:44	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec

QUESTION 6 - WHERE + SUM()

The screenshot shows MySQL Workbench with SQL File 9 open. The query is as follows:

```
28 • SELECT COUNT(*) AS Total_Transactions
29 • FROM bank_transactions;
30 • SELECT SUM(amount) AS Total_Deposit
31 • FROM bank_transactions
32 • WHERE transaction_type='Deposit';
33
```

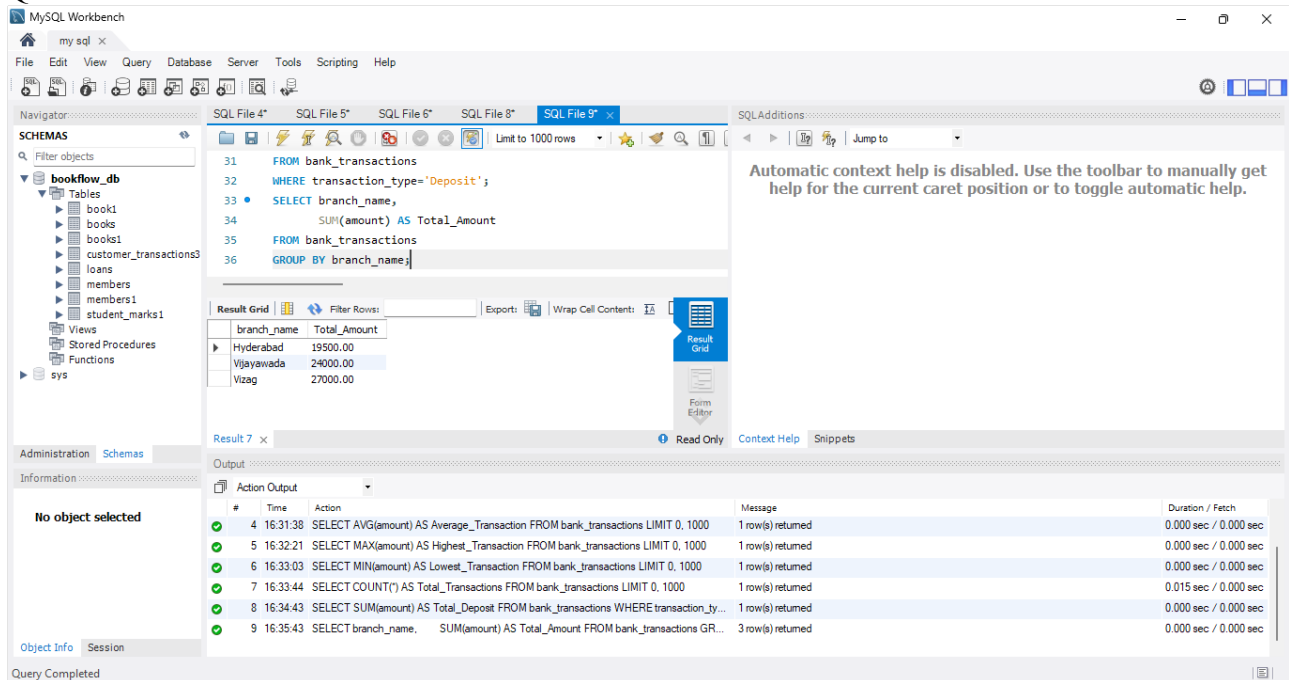
The Result Grid shows the output of the query:

	Total_Deposit
60000.00	

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
3	16:31:00	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
4	16:31:38	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	16:32:21	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:33:03	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	16:33:44	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
8	16:34:43	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_ty...	1 row(s) returned	0.000 sec / 0.000 sec

QUESTION 7 - GROUP BY



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

book1

books

books1

customer_transactions3

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

SQL File 4* SQL File 5* SQL File 6* SQL File 8* SQL File 9* x

Limit to 1000 rows

SQL

```
31 FROM bank_transactions
32 WHERE transaction_type='Deposit';
33 SELECT branch_name,
34 SUM(amount) AS Total_Amount
35 FROM bank_transactions
36 GROUP BY branch_name;
```

Result Grid

branch_name	Total_Amount
Hyderabad	19500.00
Vijayawada	24000.00
Vizag	27000.00

Result 7 x

Read Only Context Help Snippets

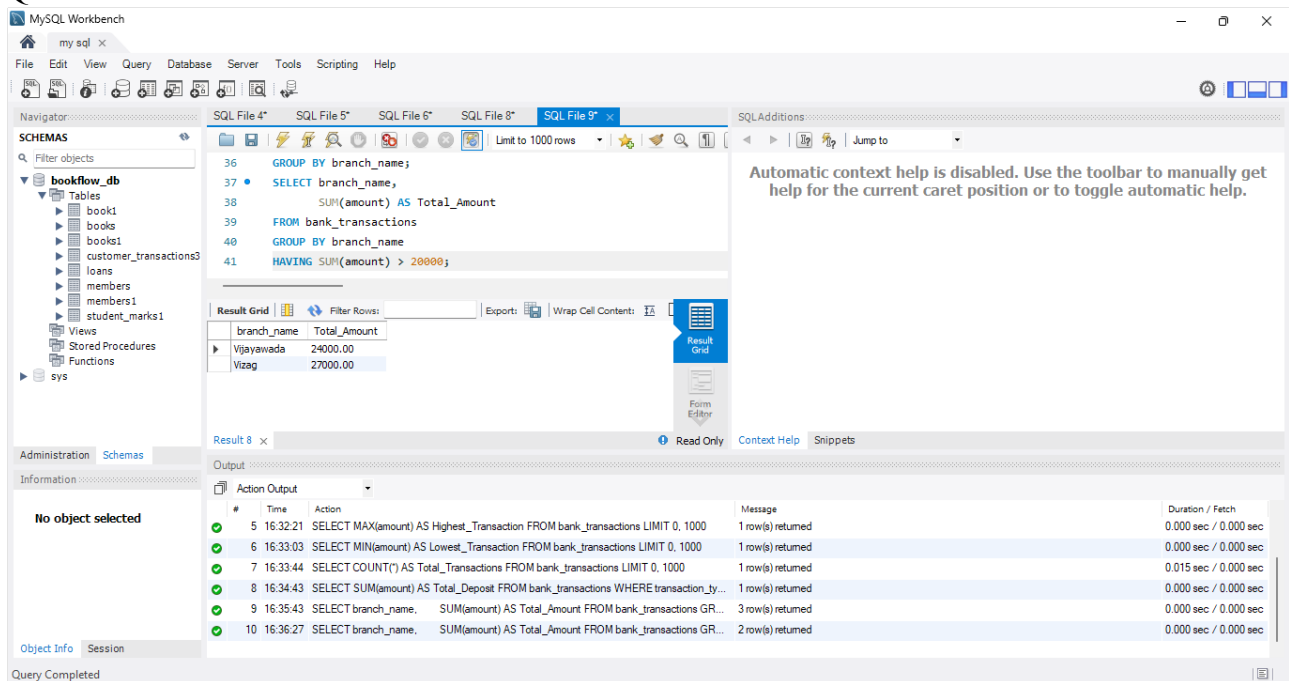
Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	16:31:38	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	16:32:21	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:33:03	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	16:33:44	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
8	16:34:43	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_ty...	1 row(s) returned	0.000 sec / 0.000 sec
9	16:35:43	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

QUESTION 8 - GROUP BY + HAVING



MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

book1

books

books1

customer_transactions3

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

SQL File 4* SQL File 5* SQL File 6* SQL File 8* SQL File 9* x

Limit to 1000 rows

SQL

```
36 GROUP BY branch_name;
37 SELECT branch_name,
38 SUM(amount) AS Total_Amount
39 FROM bank_transactions
40 GROUP BY branch_name
41 HAVING SUM(amount) > 20000;
```

Result Grid

branch_name	Total_Amount
Vijayawada	24000.00
Vizag	27000.00

Result 8 x

Read Only Context Help Snippets

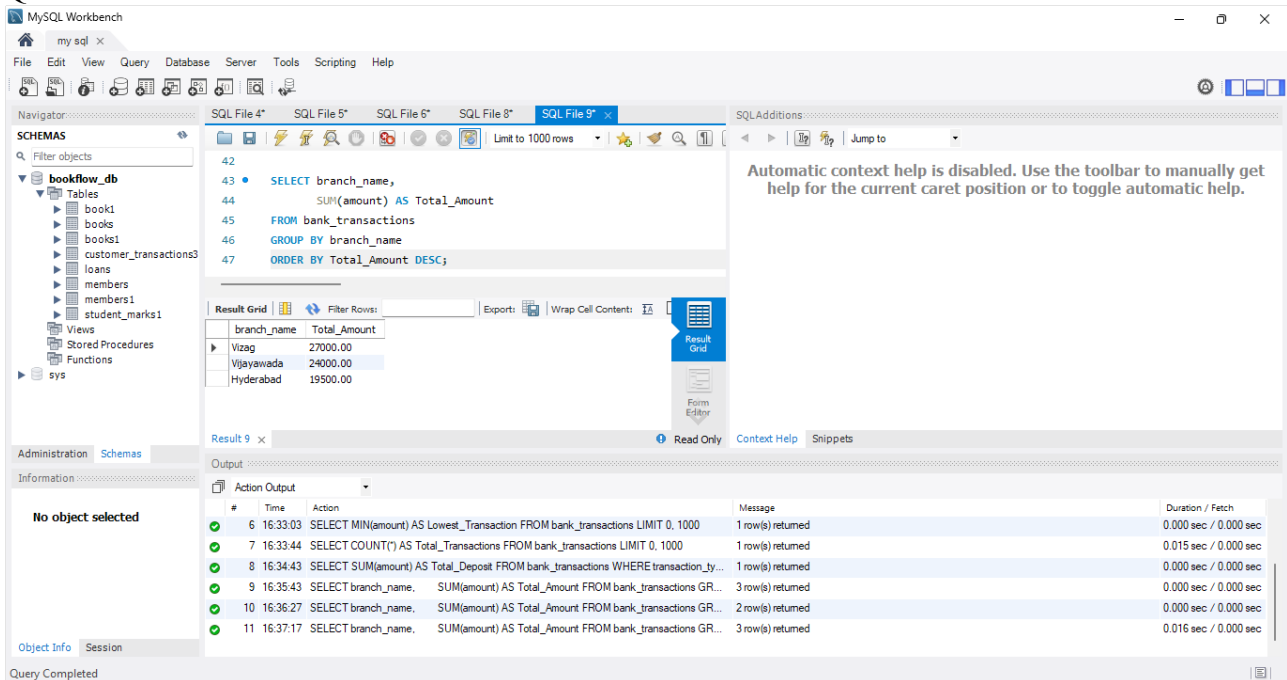
Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	16:32:21	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:33:03	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	16:33:44	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
8	16:34:43	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_ty...	1 row(s) returned	0.000 sec / 0.000 sec
9	16:35:43	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
10	16:36:27	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	2 row(s) returned	0.000 sec / 0.000 sec

Query Completed

QUESTION 9 - GROUP BY + ORDER BY



MySQL Workbench interface showing a query in SQL File 9. The query is:

```
42  
43 • SELECT branch_name,  
44      SUM(amount) AS Total_Amount  
45 FROM bank_transactions  
46 GROUP BY branch_name  
47 ORDER BY Total_Amount DESC;
```

The Result Grid shows the following data:

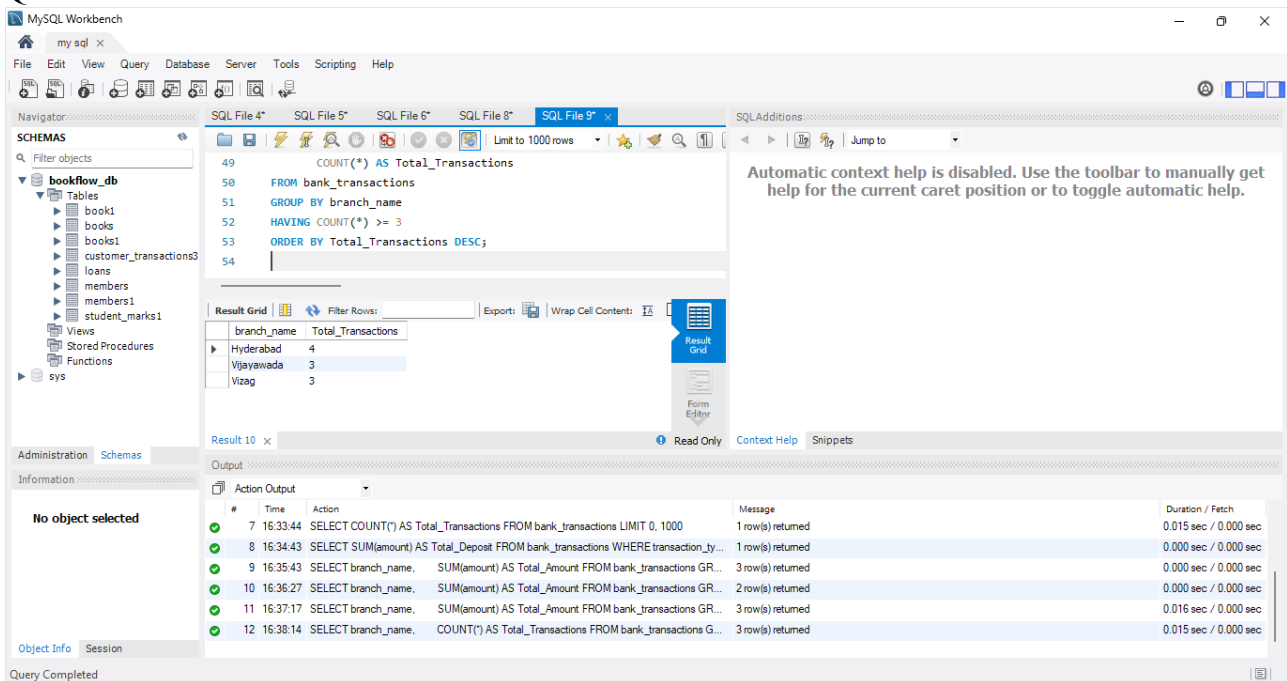
branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

The Output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
6	16:33:03	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	16:33:44	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
8	16:34:43	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_ty...	1 row(s) returned	0.000 sec / 0.000 sec
9	16:35:43	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
10	16:36:27	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	2 row(s) returned	0.000 sec / 0.000 sec
11	16:37:17	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.016 sec / 0.000 sec

Query Completed

QUESTION 10 - GROUP BY + HAVING + ORDER BY



MySQL Workbench interface showing a query in SQL File 9. The query is:

```
49 COUNT(*) AS Total_Transactions  
50 FROM bank_transactions  
51 GROUP BY branch_name  
52 HAVING COUNT(*) >= 3  
53 ORDER BY Total_Transactions DESC;  
54
```

The Result Grid shows the following data:

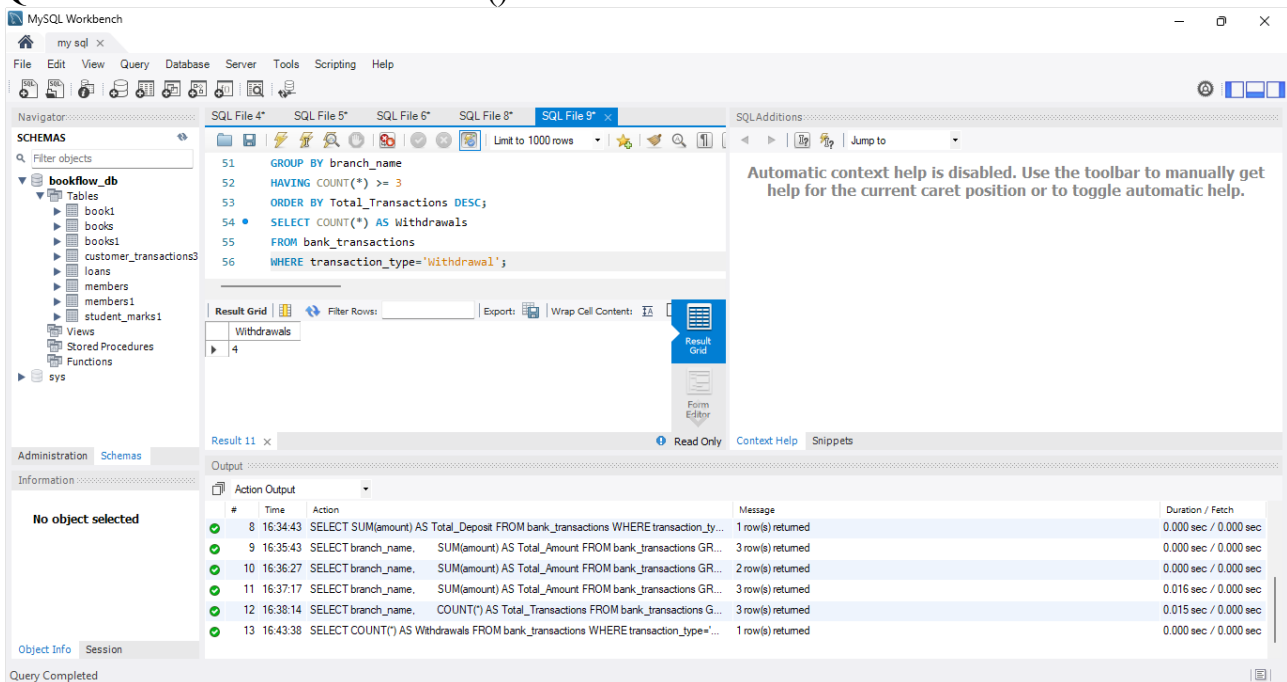
branch_name	Total_Transactions
Hyderabad	4
Vijayawada	3
Vizag	3

The Output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
7	16:33:44	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
8	16:34:43	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_ty...	1 row(s) returned	0.000 sec / 0.000 sec
9	16:35:43	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
10	16:36:27	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	2 row(s) returned	0.000 sec / 0.000 sec
11	16:37:17	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.016 sec / 0.000 sec
12	16:38:14	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	3 row(s) returned	0.015 sec / 0.000 sec

Query Completed

QUESTION 11 - WHERE + COUNT()



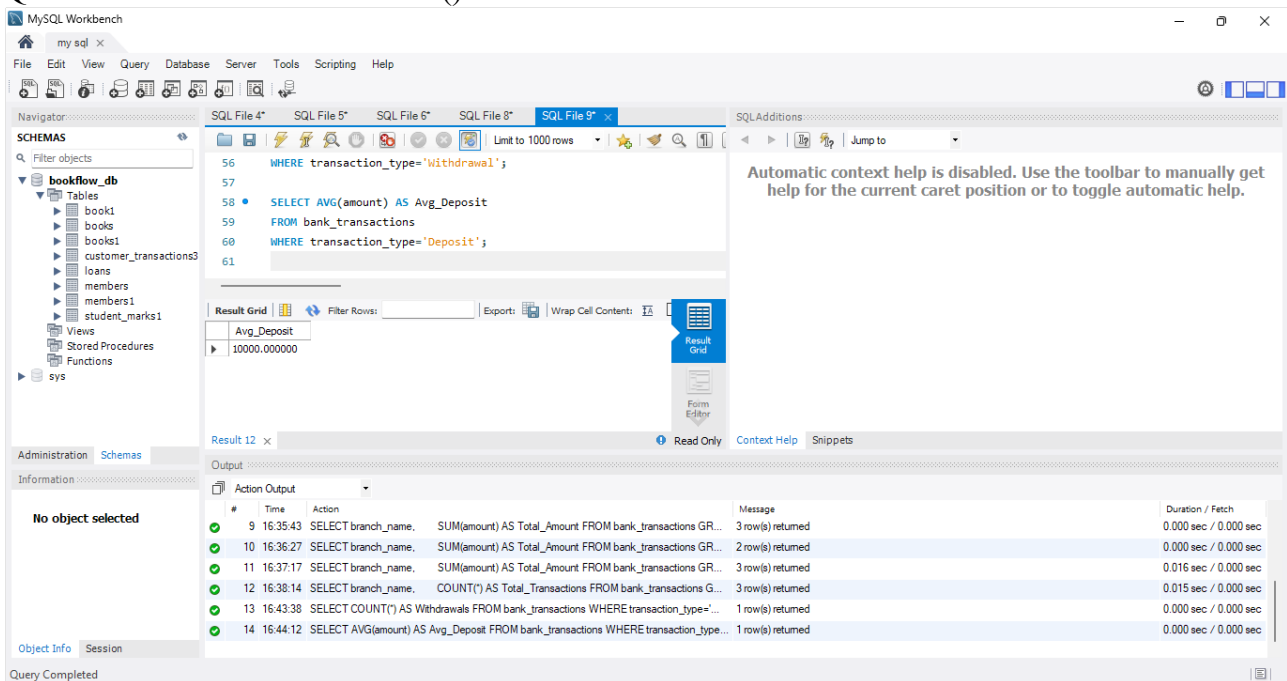
The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema with tables like 'book1', 'books', 'books1', 'customer_transactions3', 'loans', 'members', 'members1', 'student_marks1', 'Views', 'Stored Procedures', and 'Functions'. The main editor window contains the following SQL query:

```
51 GROUP BY branch_name
52 HAVING COUNT(*) >= 3
53 ORDER BY Total_Transactions DESC;
54 SELECT COUNT(*) AS Withdrawals
55 FROM bank_transactions
56 WHERE transaction_type='Withdrawal';
```

The 'Result Grid' shows a single row with the value '4' under the column 'Withdrawals'. The 'Output' pane at the bottom displays the execution log for the query, showing the time taken and the number of rows returned for each step.

#	Time	Action	Message	Duration / Fetch
8	16:34:43	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_ty...	1 row(s) returned	0.000 sec / 0.000 sec
9	16:35:43	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
10	16:36:27	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	2 row(s) returned	0.000 sec / 0.000 sec
11	16:37:17	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.016 sec / 0.000 sec
12	16:38:14	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	3 row(s) returned	0.015 sec / 0.000 sec
13	16:43:38	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type=...	1 row(s) returned	0.000 sec / 0.000 sec

QUESTION 12 - WHERE + AVG()



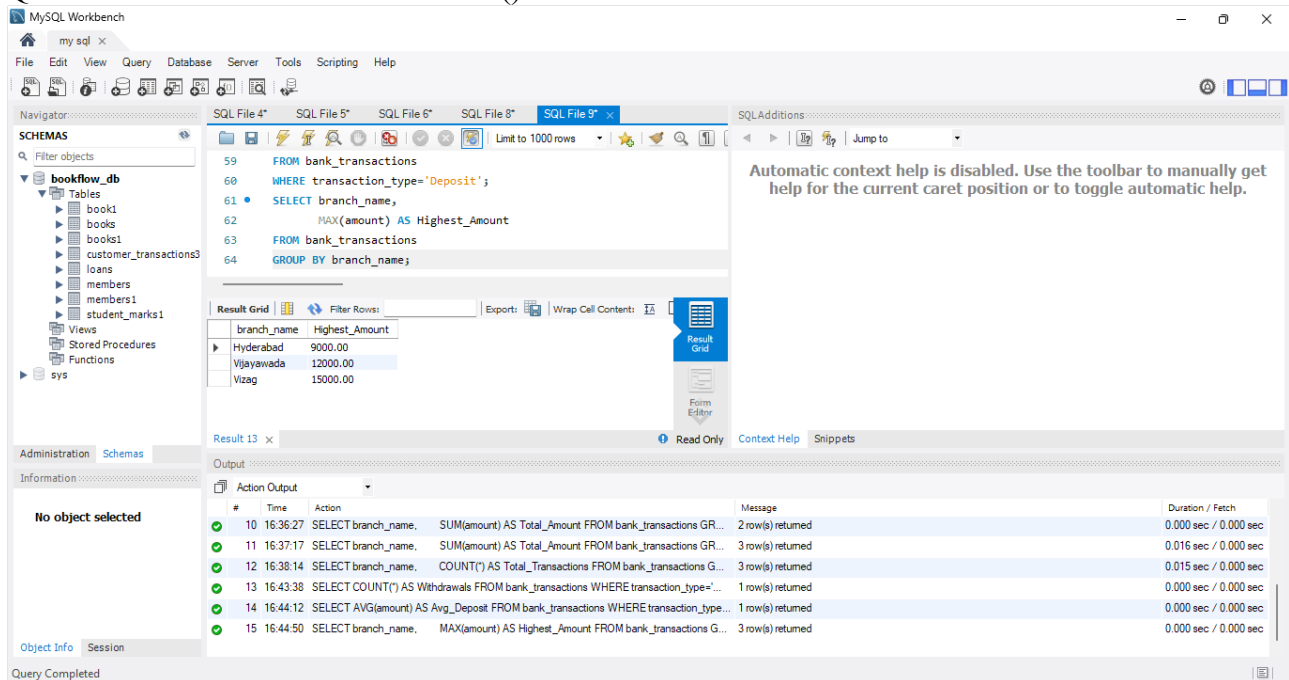
The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema with tables like 'book1', 'books', 'books1', 'customer_transactions3', 'loans', 'members', 'members1', 'student_marks1', 'Views', 'Stored Procedures', and 'Functions'. The main editor window contains the following SQL query:

```
56 WHERE transaction_type='Withdrawal';
57
58 SELECT AVG(amount) AS Avg_Deposit
59 FROM bank_transactions
60 WHERE transaction_type='Deposit';
61
```

The 'Result Grid' shows a single row with the value '10000.000000' under the column 'Avg_Deposit'. The 'Output' pane at the bottom displays the execution log for the query, showing the time taken and the number of rows returned for each step.

#	Time	Action	Message	Duration / Fetch
9	16:35:43	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
10	16:36:27	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	2 row(s) returned	0.000 sec / 0.000 sec
11	16:37:17	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.016 sec / 0.000 sec
12	16:38:14	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	3 row(s) returned	0.015 sec / 0.000 sec
13	16:43:38	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type=...	1 row(s) returned	0.000 sec / 0.000 sec
14	16:44:12	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type...	1 row(s) returned	0.000 sec / 0.000 sec

QUESTION 13 - GROUP BY + MAX()



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

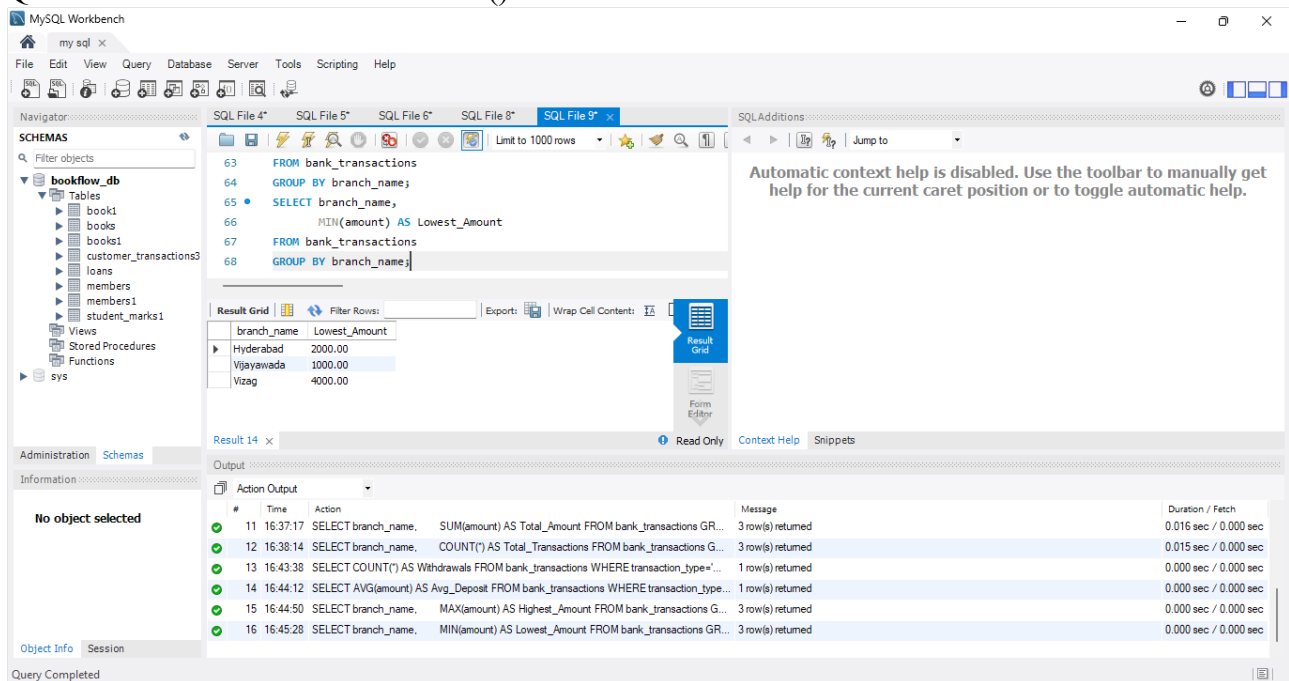
```
59 FROM bank_transactions
60 WHERE transaction_type='Deposit';
61 SELECT branch_name,
62        MAX(amount) AS Highest_Amount
63 FROM bank_transactions
64 GROUP BY branch_name;
```

The result grid displays the following data:

branch_name	Highest_Amount
Hyderabad	9000.00
Vijayawada	12000.00
Vizag	15000.00

The output pane shows the execution log for the query, indicating that 3 rows were returned.

QUESTION 14 - GROUP BY + MIN()



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

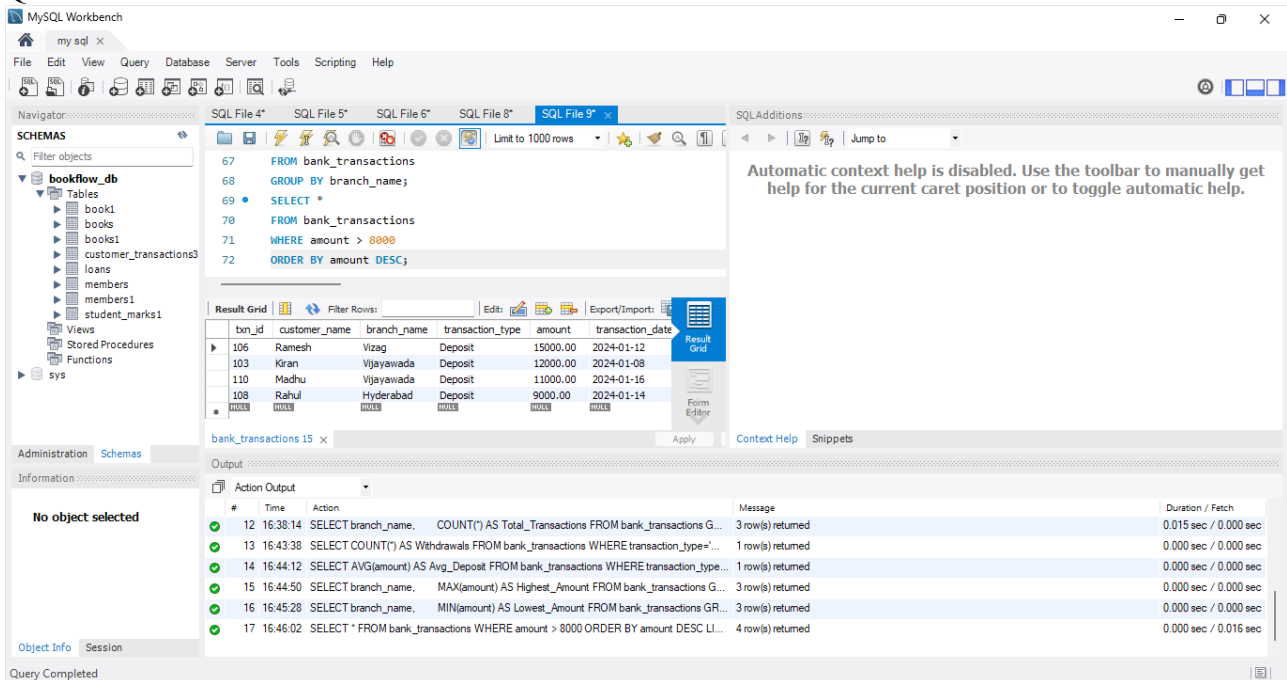
```
63 FROM bank_transactions
64 GROUP BY branch_name;
65 SELECT branch_name,
66        MIN(amount) AS Lowest_Amount
67 FROM bank_transactions
68 GROUP BY branch_name;
```

The result grid displays the following data:

branch_name	Lowest_Amount
Hyderabad	2000.00
Vijayawada	1000.00
Vizag	4000.00

The output pane shows the execution log for the query, indicating that 3 rows were returned.

QUESTION 15 - WHERE + ORDER BY



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' database with various tables and views. The central SQL editor contains a query that filters transactions by amount and orders them by amount in descending order. The 'Result Grid' shows the output of this query, listing transaction details for five different branches. The 'Output' pane at the bottom shows the execution progress of the query.

```
67 FROM bank_transactions
68 GROUP BY branch_name;
69 SELECT *
70 FROM bank_transactions
71 WHERE amount > 8000
72 ORDER BY amount DESC;
```

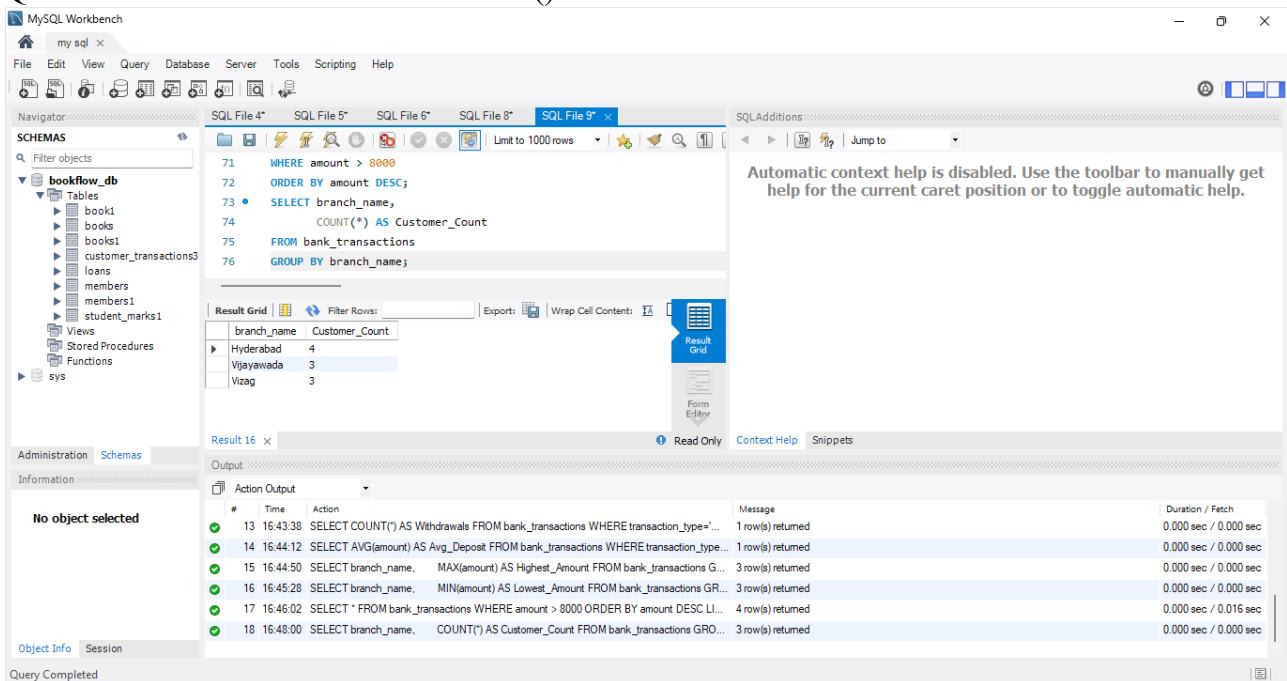
txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14

bank_transactions 15 x

Output

#	Time	Action	Message	Duration / Fetch
12	16:38:14	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	3 row(s) returned	0.015 sec / 0.000 sec
13	16:43:38	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type=...	1 row(s) returned	0.000 sec / 0.000 sec
14	16:44:12	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type...	1 row(s) returned	0.000 sec / 0.000 sec
15	16:44:50	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions G...	3 row(s) returned	0.000 sec / 0.000 sec
16	16:45:28	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
17	16:46:02	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LI...	4 row(s) returned	0.000 sec / 0.016 sec

QUESTION 16 - GROUP BY + COUNT()



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' database. The central SQL editor contains a query that groups transactions by branch name and counts the number of transactions for each branch. The 'Result Grid' shows the output of this query, listing the branch names and their corresponding transaction counts. The 'Output' pane at the bottom shows the execution progress of the query.

```
71 WHERE amount > 8000
72 ORDER BY amount DESC;
73 SELECT branch_name,
74 COUNT(*) AS Customer_Count
75 FROM bank_transactions
76 GROUP BY branch_name;
```

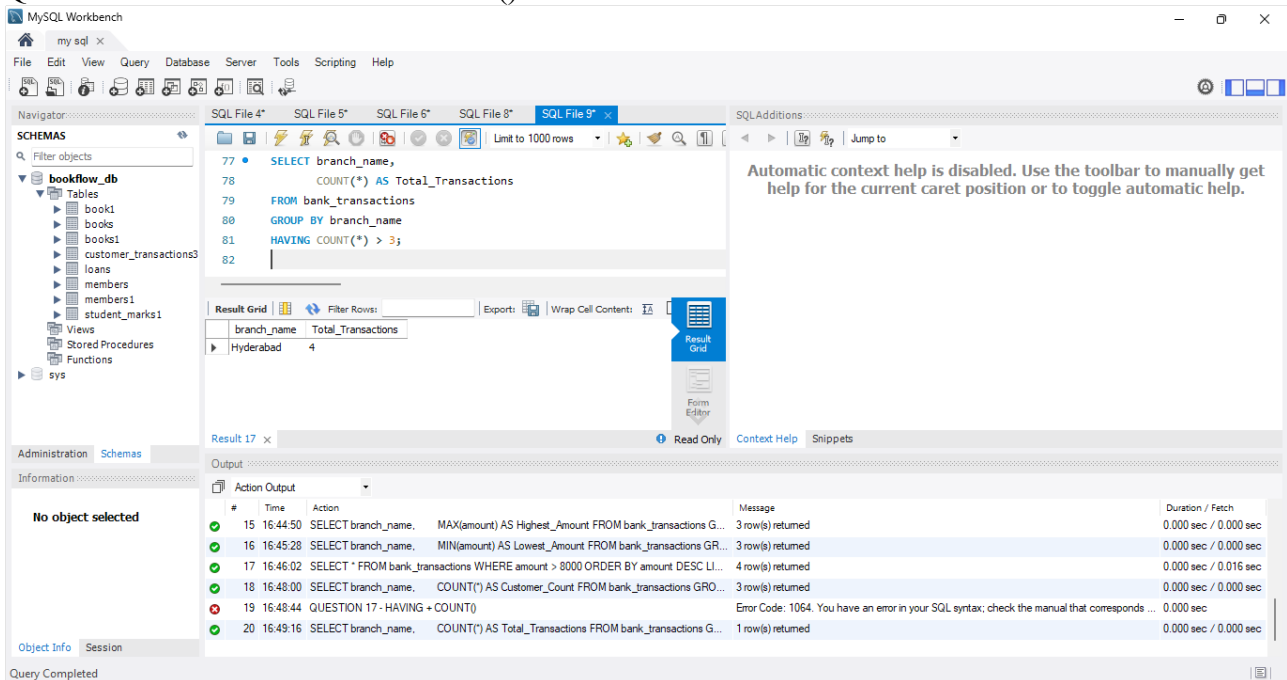
branch_name	Customer_Count
Hyderabad	4
Vijayawada	3
Vizag	3

Result 16 x

Output

#	Time	Action	Message	Duration / Fetch
13	16:43:38	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type=...	1 row(s) returned	0.000 sec / 0.000 sec
14	16:44:12	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type...	1 row(s) returned	0.000 sec / 0.000 sec
15	16:44:50	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions G...	3 row(s) returned	0.000 sec / 0.000 sec
16	16:45:28	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
17	16:46:02	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LI...	4 row(s) returned	0.000 sec / 0.016 sec
18	16:48:00	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec

QUESTION 17 - HAVING + COUNT()



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema with tables like 'book1', 'books', 'books1', 'customer_transactions3', 'loans', 'members', 'members1', 'student_marks1', 'Views', 'Stored Procedures', and 'Functions'. The main editor window shows a SQL query in 'SQL File 9':

```
77 SELECT branch_name,  
78 COUNT(*) AS Total_Transactions  
79 FROM bank_transactions  
80 GROUP BY branch_name  
81 HAVING COUNT(*) > 3;  
82
```

The 'Result Grid' below the query shows one row of results:

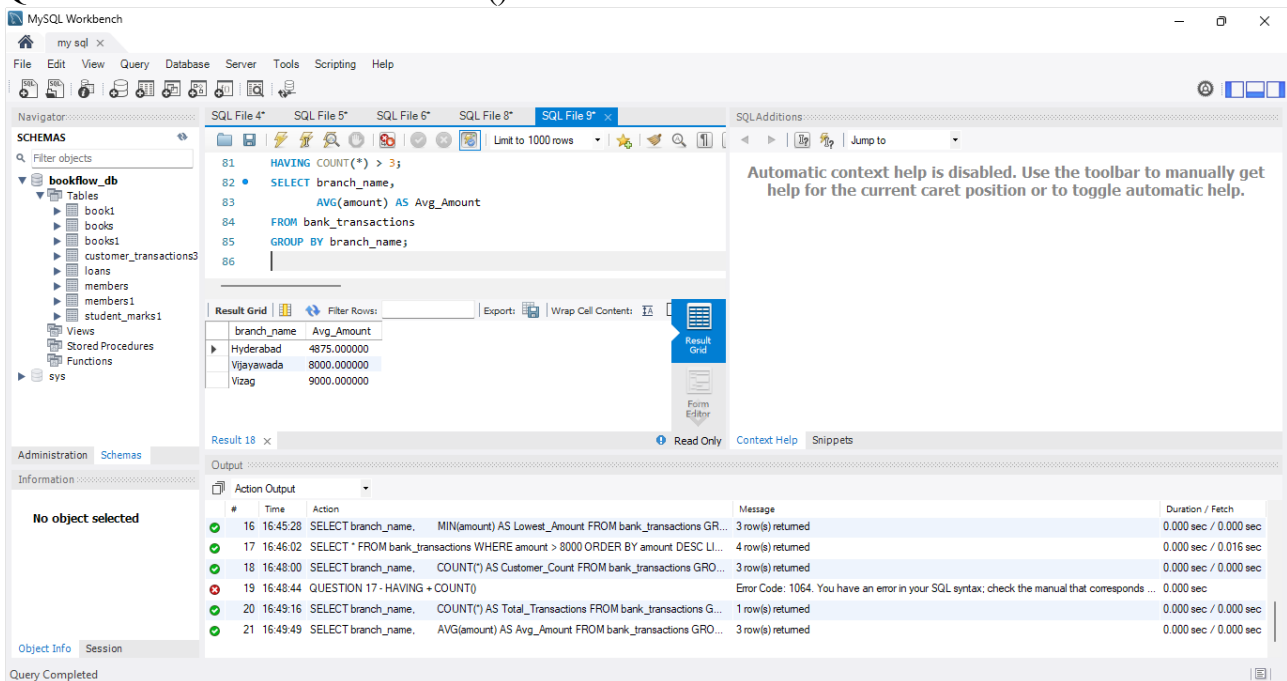
branch_name	Total_Transactions
Hyderabad	4

The 'Output' pane at the bottom shows the execution log. The last entry is an error:

#	Time	Action	Message	Duration / Fetch
15	16:44:50	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions G...	3 row(s) returned	0.000 sec / 0.000 sec
16	16:45:28	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
17	16:46:02	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LI...	4 row(s) returned	0.000 sec / 0.016 sec
18	16:48:00	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec
19	16:48:44	QUESTION 17 - HAVING + COUNT()	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
20	16:49:16	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	1 row(s) returned	0.000 sec / 0.000 sec

The error message indicates a syntax error in the HAVING clause. The right sidebar contains a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

QUESTION 18 - GROUP BY + AVG()



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema. The main editor window shows a SQL query in 'SQL File 9':

```
81 HAVING COUNT(*) > 3;  
82 SELECT branch_name,  
83 AVG(amount) AS Avg_Amount  
84 FROM bank_transactions  
85 GROUP BY branch_name;  
86
```

The 'Result Grid' below the query shows three rows of results:

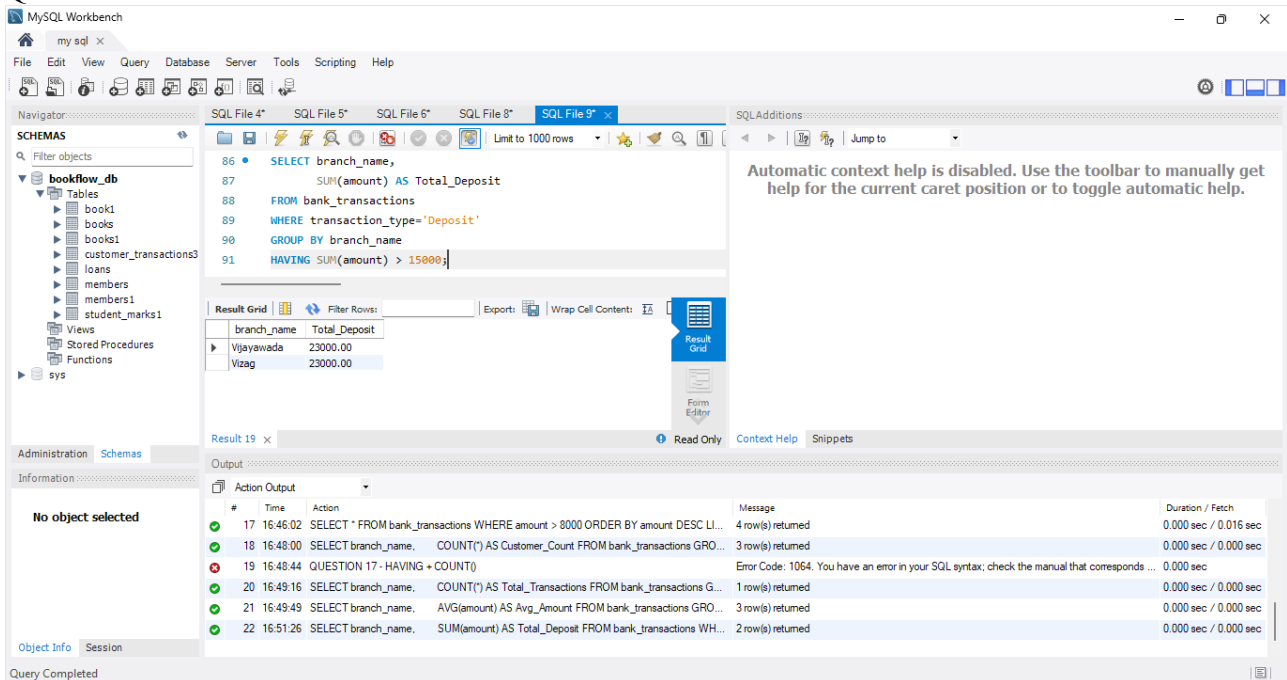
branch_name	Avg_Amount
Hyderabad	4875.000000
Vijayawada	8000.000000
Vizag	9000.000000

The 'Output' pane at the bottom shows the execution log. The last entry is an error:

#	Time	Action	Message	Duration / Fetch
16	16:45:28	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
17	16:46:02	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LI...	4 row(s) returned	0.000 sec / 0.016 sec
18	16:48:00	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec
19	16:48:44	QUESTION 17 - HAVING + COUNT()	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
20	16:49:16	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	1 row(s) returned	0.000 sec / 0.000 sec
21	16:49:49	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec

The error message indicates a syntax error in the HAVING clause. The right sidebar contains a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

QUESTION 19 - WHERE + GROUP BY + HAVING



MySQL Workbench interface showing a SQL query in SQL File 9. The query is:

```
86 SELECT branch_name,  
87 SUM(amount) AS Total_Deposit  
88 FROM bank_transactions  
89 WHERE transaction_type='Deposit'  
90 GROUP BY branch_name  
91 HAVING SUM(amount) > 15000;
```

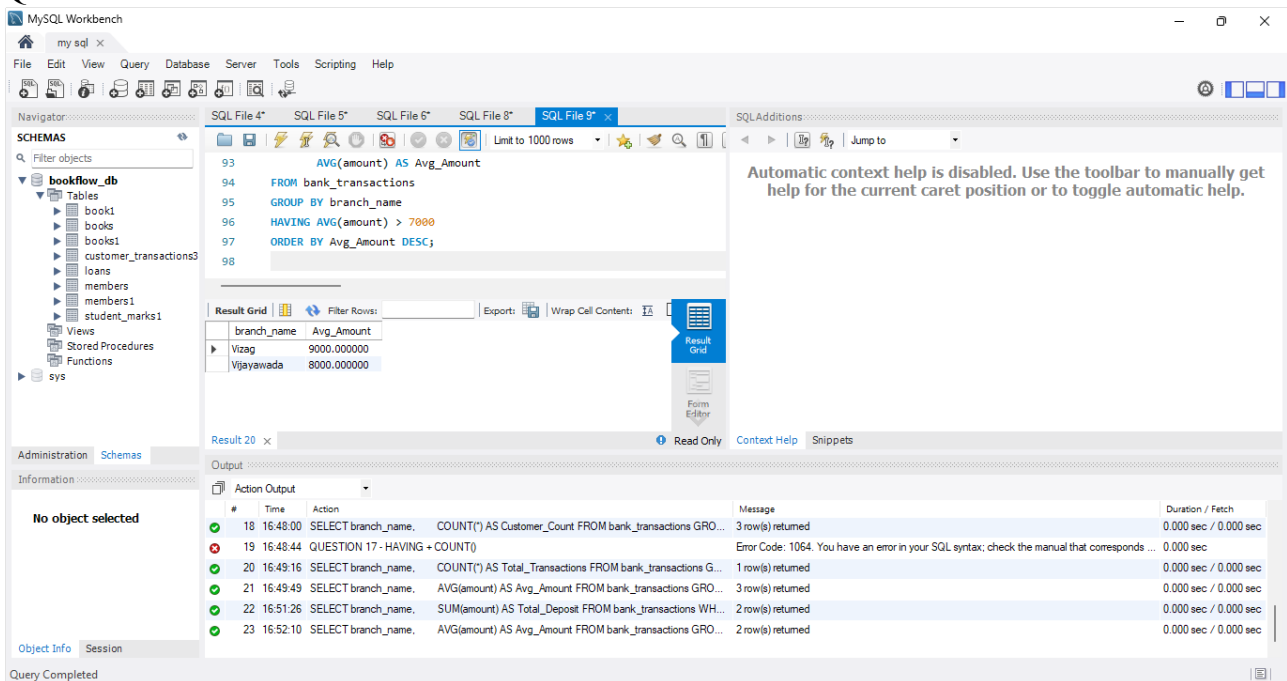
The result grid shows two rows:

branch_name	Total_Deposit
Vijayawada	23000.00
Vizag	23000.00

The output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
17	16:46:02	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LI...	4 row(s) returned	0.000 sec / 0.016 sec
18	16:48:00	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec
19	16:48:44	QUESTION 17 - HAVING + COUNT()	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
20	16:49:16	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	1 row(s) returned	0.000 sec / 0.000 sec
21	16:49:49	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec
22	16:51:26	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WH...	2 row(s) returned	0.000 sec / 0.000 sec

QUESTION 20 - GROUP BY + HAVING + ORDER BY



MySQL Workbench interface showing a SQL query in SQL File 9. The query is:

```
93 AVG(amount) AS Avg_Amount  
94 FROM bank_transactions  
95 GROUP BY branch_name  
96 HAVING AVG(amount) > 7000  
97 ORDER BY Avg_Amount DESC;  
98
```

The result grid shows two rows:

branch_name	Avg_Amount
Vizag	9000.000000
Vijayawada	8000.000000

The output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
18	16:48:00	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec
19	16:48:44	QUESTION 17 - HAVING + COUNT()	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
20	16:49:16	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	1 row(s) returned	0.000 sec / 0.000 sec
21	16:49:49	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec
22	16:51:26	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WH...	2 row(s) returned	0.000 sec / 0.000 sec
23	16:52:10	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GRO...	2 row(s) returned	0.000 sec / 0.000 sec

FINAL OUTPUTS:

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

- Tables
 - book1
 - books
 - books1
 - customer_transactions3
 - loans
 - members
 - members1
 - student_marks1
- Views
- Stored Procedures
- Functions

sys

Administration Schemas

Information

No object selected

Object Info Session

Query Completed

SQL File 4* SQL File 5* SQL File 6* SQL File 8* SQL File 9* x SQLAdditions

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	16:28:28	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VAR...	0 row(s) affected	0.062 sec
2	16:30:06	INSERT INTO bank_transactions VALUES (101,'Sithika','Hyderabad','Deposit',5000,2024-0...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	16:31:00	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
4	16:31:38	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	16:32:21	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	16:33:03	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
7	16:33:44	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.000 sec
8	16:34:43	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_ty...	1 row(s) returned	0.000 sec / 0.000 sec
9	16:35:43	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
10	16:36:27	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	2 row(s) returned	0.000 sec / 0.000 sec
11	16:37:17	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GR...	3 row(s) returned	0.016 sec / 0.000 sec
12	16:38:14	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	3 row(s) returned	0.015 sec / 0.000 sec
13	16:43:38	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type=...	1 row(s) returned	0.000 sec / 0.000 sec
14	16:44:12	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type...	1 row(s) returned	0.000 sec / 0.000 sec
15	16:44:50	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions G...	3 row(s) returned	0.000 sec / 0.000 sec
16	16:45:28	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GR...	3 row(s) returned	0.000 sec / 0.000 sec
17	16:46:02	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LI...	4 row(s) returned	0.000 sec / 0.016 sec
18	16:48:00	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec
19	16:48:44	QUESTION 17 - HAVING + COUNT()	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
20	16:49:16	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions G...	1 row(s) returned	0.000 sec / 0.000 sec
21	16:49:49	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GRO...	3 row(s) returned	0.000 sec / 0.000 sec
22	16:51:26	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WH...	2 row(s) returned	0.000 sec / 0.000 sec